ref input, SimPoint estimates — small-4way

L1D 16 sets x 4, L2 64 sets x 8 · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

36 of 36 runs completed, 0 running, 0 pending; **0 remaining**.

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	1/1	20.65%	47.95%	56.30%	20.67%	48.00%	56.39%
parest	1/1	0.16%	0.43%	0.62%	0.64%	1.19%	7.86%
deepsjeng	1/1	2.92%	9.17%	16.25%	8.11%	23.58%	49.72%
leela	1/1	0.38%	1.36%	1.80%	6.07%	15.90%	45.53%
namd	1/1	3.27%	8.21%	17.87%	3.27%	8.22%	17.90%
lbm	1/1	10.48%	32.28%	44.97%	10.48%	32.28%	44.97%
nab	1/1	5.20%	15.37%	34.02%	6.75%	18.74%	49.89%
xz_cld	1/1	3.38%	12.11%	22.88%	5.08%	17.35%	36.71%
xz_cpu2006docs	1/1	1.47%	5.78%	23.30%	2.03%	7.38%	37.90%
xz_combined	1/1	2.85%	10.66%	26.32%	4.17%	14.65%	42.32%
imagick	1/1	0.23%	0.12%	3.44%	5.33%	12.37%	46.11%
blender	1/1	0.38%	0.88%	3.23%	3.72%	6.96%	42.51%
povray	1/1	0.27%	0.46%	0.95%	14.26%	22.18%	55.24%
omnetpp	1/1	0.00%	0.00%	0.00%	12.75%	24.67%	48.13%
xalancbmk	1/1	0.13%	0.26%	0.80%	4.92%	8.32%	39.84%
x264_pass1	1/1	7.07%	15.02%	38.53%	10.18%	21.51%	55.90%
x264_pass2	1/1	6.25%	13.38%	40.16%	8.63%	18.23%	56.21%
x264_seek	1/1	6.41%	13.66%	41.07%	8.82%	18.56%	57.26%
gcc_pp_O3	1/1	1.23%	2.73%	4.48%	16.24%	33.66%	64.23%
gcc_pp_O2	1/1	1.25%	2.75%	4.47%	16.60%	34.40%	63.73%
gcc_smaller	1/1	1.09%	2.19%	4.61%	13.76%	25.29%	64.09%
gcc_ref32_O5	1/1	0.91%	1.84%	3.40%	15.53%	30.63%	59.73%
gcc_ref32_O3	1/1	0.68%	1.85%	1.06%	11.25%	27.66%	19.94%
mcf	1/1	1.40%	2.06%	7.74%	2.33%	3.60%	12.43%
perlbench_checkspam	1/1	7.98%	9.83%	29.42%	22.33%	36.24%	69.19%
perlbench_diffmail	1/1	1.96%	4.19%	4.64%	17.81%	31.25%	54.38%
perlbench_splitmail	1/1	4.16%	4.19%	15.82%	23.27%	35.82%	68.04%
bwaves_1	1/1	2.51%	6.24%	18.43%	2.51%	6.24%	18.43%
bwaves_2	1/1	2.59%	6.43%	18.99%	2.59%	6.43%	18.99%
bwaves_3	1/1	2.51%	6.23%	18.45%	2.51%	6.23%	18.45%
bwaves_4	1/1	2.56%	6.32%	19.44%	2.56%	6.32%	19.44%
exchange2	1/1	5.23%	10.55%	12.68%	6.08%	12.17%	14.84%
fotonik3d	1/1	3.36%	9.32%	9.25%	3.38%	9.37%	9.40%
roms	1/1	4.37%	13.83%	8.97%	4.44%	13.96%	9.48%
wrf	1/1	0.91%	3.38%	2.59%	1.52%	4.76%	7.57%
cam4	1/1	3.14%	11.02%	11.03%	4.99%	15.77%	21.23%
pooled		3.78%	9.06%	17.53%	7.49%	16.44%	39.26%
mean		3.31%	8.11%	15.78%	8.49%	18.33%	39.00%

Miss rates

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.63%	5.18%	33.25%	4.30%	13.06%	64.99%	12.26%	9.77%
parest	done	10.38%	2.36%	10.98%	1.67%	5.06%	70.85%	3.14%	27.35%
deepsjeng	done	8.81%	2.10%	11.44%	7.45%	4.16%	6.35%	2.23%	14.14%
leela	done	4.94%	0.51%	5.49%	0.00%	3.55%	23.48%	0.03%	21.59%
namd	done	11.58%	2.40%	14.75%	1.26%	2.43%	27.50%	7.59%	16.11%
lbm	done	17.63%	0.01%	6.46%	0.12%	75.48%	97.01%	0.06%	12.21%
nab	done	5.40%	0.13%	7.58%	0.69%	0.43%	59.18%	9.30%	1.65%
xz_cld	done	10.66%	2.04%	15.30%	1.51%	2.95%	35.09%	0.03%	0.26%
xz_cpu2006docs	done	9.40%	2.20%	10.33%	1.90%	6.32%	39.60%	4.81%	6.67%
xz_combined	done	9.72%	1.98%	12.37%	1.30%	4.34%	24.47%	0.01%	1.29%
imagick	done	2.18%	3.93%	2.77%	0.03%	0.12%	96.60%	57.37%	19.19%
blender	done	8.00%	2.67%	8.76%	1.42%	3.47%	18.31%	1.21%	46.53%
povray	done	9.16%	7.21%	11.34%	18.52%	3.61%	45.94%	51.80%	63.63%
omnetpp	done	16.31%	0.08%	21.74%	1.50%	5.71%	28.08%	50.69%	2.63%
xalancbmk	done	17.12%	11.80%	19.71%	22.71%	4.54%	70.02%	6.11%	5.53%
x264_pass1	done	2.42%	0.13%	2.87%	1.00%	3.14%	28.58%	6.19%	15.50%
x264_pass2	done	1.95%	0.19%	2.37%	0.78%	2.00%	35.53%	0.85%	18.84%
x264_seek	done	1.96%	0.16%	2.41%	0.91%	1.93%	31.56%	1.11%	13.54%
gcc_pp_O3	done	8.74%	1.30%	11.63%	4.77%	3.64%	41.84%	5.87%	12.58%
gcc_pp_O2	done	8.89%	1.19%	11.84%	4.31%	3.90%	38.46%	6.57%	11.89%
gcc_smaller	done	10.48%	2.05%	13.46%	3.71%	3.44%	47.59%	3.59%	10.17%
gcc_ref32_O5	done	8.86%	1.68%	11.62%	4.07%	3.81%	50.61%	7.41%	12.73%
gcc_ref32_O3	done	10.65%	2.48%	12.31%	5.12%	9.62%	72.66%	4.57%	18.30%
mcf	done	21.53%	6.09%	27.47%	6.44%	4.64%	74.18%	3.56%	19.24%
perlbench_checkspam	done	2.50%	0.10%	3.92%	0.43%	1.13%	28.16%	2.52%	8.95%
perlbench_diffmail	done	4.51%	0.13%	6.17%	4.56%	1.79%	37.92%	0.37%	12.19%
perlbench_splitmail	done	8.88%	0.03%	12.58%	7.27%	2.96%	8.50%	2.60%	0.41%
bwaves_1	done	44.12%	6.22%	48.24%	0.10%	36.83%	23.63%	3.19%	0.58%
bwaves_2	done	43.90%	6.02%	48.01%	0.12%	37.89%	23.88%	3.03%	0.66%
bwaves_3	done	43.19%	6.11%	47.21%	0.00%	36.20%	24.11%	3.53%	67.82%
bwaves_4	done	43.28%	5.84%	47.31%	0.00%	36.77%	23.87%	2.64%	38.55%
exchange2	done	1.67%	1.41%	2.15%	2.24%	0.96%	0.94%	0.09%	0.40%
fotonik3d	done	10.25%	4.03%	14.24%	0.04%	0.18%	66.95%	1.93%	6.94%
roms	done	15.13%	2.31%	18.45%	0.52%	11.85%	53.37%	13.34%	13.66%
wrf	done	11.27%	6.63%	11.52%	4.40%	11.13%	33.77%	8.59%	11.72%
cam4	done	10.69%	2.04%	11.57%	2.35%	12.15%	55.53%	8.09%	25.31%
pooled		10.79%	3.07%	13.38%	2.69%	6.61%	47.07%	10.11%	11.85%
mean		13.11%	2.80%	15.55%	3.26%	10.03%	41.92%	8.23%	15.79%

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save

column	formula	aciklama
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.